1.55 Discrete Random Variables

Worked Solutions

1)

$\mathrm{P}(2 \leq X < 5) = 0.13 + 0.29 + 0.30 = 0.72$

2)

x	2	4	6	8
$\mathrm{P}(X = x)$	$\frac{11}{50}$	$\frac{6}{25}$	$\frac{13}{50}$	$\frac{7}{25}$

3)

$\mathrm{P}(X > 4) = \mathrm{P}(X = 5) + \mathrm{P}(X = 7) = \frac{5+6}{40} + \frac{7+6}{40} = \frac{3}{5}$

4)

$0.05 + 0.1 + k + 0.35 + 0.32 = 1 \Rightarrow k + 0.82 = 1$

$\Rightarrow k = 0.18$

5)

$k + 3k + 5k + k + k = 1 \Rightarrow 11k = 1 \Rightarrow k = \frac{1}{11}$

6 a)

$k(10-2) + k(10-4) + k(10-6) + k(10-8) = 1$

$\Rightarrow 20k = 1 \Rightarrow k = \frac{1}{20}$

6 b)

$\mathrm{P}(X \geq 6) = \frac{1}{20}(10-6) + \frac{1}{20}(10-8) = \frac{3}{10}$

7)

$\frac{1+k}{10+10} + \frac{2+k}{20+10} + \frac{3+k}{30+10} = 1$

$\Rightarrow \frac{1+k}{20} + \frac{2+k}{30} + \frac{3+k}{40} = 1$

$\Rightarrow \frac{6+6k}{120} + \frac{8+4k}{120} + \frac{9+3k}{120} = 1$

$\Rightarrow 6 + 6k + 8 + 4k + 9 + 3k = 120$

$\Rightarrow 13k = 97 \Rightarrow k = \frac{97}{13}$

8)

$\mathrm{P}(1 < X \leq 4) = 0.1 + 0.25 + 0.3 = 0.65$

9)

$\mathrm{P}(X \geq 4) = 0.2 + 0.25 = 0.45$

$\mathrm{P}(\text{both values} \geq 4) = 0.45^2 = 0.2025$

10)

$\mathrm{P}(1,1) + \mathrm{P}(2,2) + \mathrm{P}(3,3) = 0.71^2 + 0.19^2 + 0.1^2 = 0.5502$

11)

$\mathrm{P}(2,1) + \mathrm{P}(3,1) + \mathrm{P}(3,2)$

$= 0.09 \times 0.9 + 0.01 \times 0.9 + 0.01 \times 0.09 = 0.0909$

12)

$\mathrm{P}(1,2) + \mathrm{P}(1,3) + \mathrm{P}(3,2) + \mathrm{P}(3,3)$

$= 0.24 \times 0.36 + 0.24 \times 0.4 + 0.4 \times 0.36 + 0.4^2$

$= 0.4864$

13)

$\mathrm{P}(5,5) + \mathrm{P}(5,8) + \mathrm{P}(5,11) + \mathrm{P}(8,5) + \mathrm{P}(8,8) + \mathrm{P}(11,5)$

$= 0.15^2 + 0.15 \times 0.25 + 0.15 \times 0.6 + 0.25 \times 0.15 + 0.25^2 + 0.6 \times 0.15 = 0.34$

14)

$\mathrm{P}(1,3) + \mathrm{P}(3,1) + \mathrm{P}(2,2)$

$= 0.02 \times 0.26 + 0.26 \times 0.02 + 0.14 \times 0.14 = 0.03$

15)

$\mathrm{P}(0,0) + \mathrm{P}(0,1) + \mathrm{P}(1,0) + \mathrm{P}(0,2) + \mathrm{P}(2,0)$

$= 0.33^2 + 2 \times 0.33 \times 0.22 + 2 \times 0.33 \times 0.45$

$= 0.5511$

16)

$\mathrm{P}(3,3) + \mathrm{P}(4,4) + \mathrm{P}(6,6) + \mathrm{P}(7,7)$

$= \left(\frac{3}{20}\right)^2 + \left(\frac{4}{20}\right)^2 + \left(\frac{6}{20}\right)^2 + \left(\frac{7}{20}\right)^2 = \frac{11}{40}$

17)

x	1	2	3	4
$\mathrm{P}(X = x)$	$\frac{1}{7}$	$\frac{3}{14}$	$\frac{2}{7}$	$\frac{5}{14}$

$\mathrm{P}(3, 1+1) + \mathrm{P}(4, 1+1) + \mathrm{P}(4, 1+2) + \mathrm{P}(4, 2+1)$

$= \frac{2}{7} \times \frac{1}{7} \times \frac{1}{7} + \frac{5}{14} \times \frac{1}{7} \times \frac{1}{7} + \frac{5}{14} \times \frac{1}{7} \times \frac{3}{14} + \frac{5}{14} \times \frac{3}{14} \times \frac{1}{7}$

$= \frac{2}{343} + \frac{5}{686} + \frac{15}{1372} + \frac{15}{1372} = \frac{12}{343}$

18)

x	1	2	4	5
$\mathrm{P}(X=x)$	0.2	0.3	0.2	0.3

19)

x	4	5	9	10
$\mathrm{P}(X=x)$	0.32	0.18	0.32	0.18

20)

x	2	3	4	5	6
$\mathrm{P}(X=x)$	0.11	k	p	0.11	k

$\sum \mathrm{P}(X=x)=1 \Rightarrow 0.11+k+p+0.11+k=1$

$\Rightarrow 2k+p=0.78$

$\mathrm{P}(X=4)=\mathrm{P}(X=3)+\mathrm{P}(X=5) \Rightarrow p=k+0.11$

$\Rightarrow -k+p=0.11$

$\therefore k=\frac{67}{300}$ and $p=\frac{1}{3}$

x	2	3	4	5	6
$\mathrm{P}(X=x)$	0.11	$\frac{67}{300}$	$\frac{1}{3}$	0.11	$\frac{67}{300}$

21 a)

$\mathrm{P}(X=2)=\frac{1}{5}\times\frac{2^{2-1}}{3^{2-1}}=\frac{2}{15}$

21 b)

$\frac{1}{5}+\frac{2}{15}+k+k=1 \Rightarrow 2k+\frac{1}{3}=1 \Rightarrow k=\frac{1}{3}$

21 c)

$\mathrm{P}(2\leq X\leq 3)=\frac{2}{15}+\frac{1}{3}=\frac{7}{15}$

22 a)

$\mathrm{P}(X=2)=a\times 2^2=4a$

22 b)

$\mathrm{P}(X=2)=\mathrm{P}(X\geq 3) \Rightarrow 4a=3b+0.05+4b+0.05$

$\Rightarrow 4a-7b=0.1$

$a+4a+3b+0.05+4b+0.05=1$

$\Rightarrow 5a+7b=0.9$

$a=\frac{1}{9}$ and $b=\frac{31}{630}$

23 a)

$12a + 9a + 6a + 3a + 5b + 6b = 1$

$\Rightarrow 30a + 11b = 1$

23 b)

$\mathrm{P}(X \geq 4) = \frac{1}{4} \Rightarrow 3a + 5b + 6b = \frac{1}{4} \Rightarrow 3a + 11b = \frac{1}{4}$

$a = \frac{1}{36}$ and $b = \frac{1}{66}$

24)

$3k^2 + 3k + \frac{9}{2}k^2 + \frac{1}{2}k = 1 \Rightarrow \frac{15}{2}k^2 + \frac{7}{2}k - 1 = 0$

$\Rightarrow 15k^2 + 7k - 2 = 0 \Rightarrow (5k - 1)(3k + 2) = 0$

$\Rightarrow k = \frac{1}{5}$ as $k \neq -\frac{2}{3}$

$\mathrm{P}(X > 1) = 1 - 3 \times \left(\frac{1}{5}\right)^2 = \frac{22}{25}$

25)

$k + k + k + k + \frac{1}{3}k + \frac{1}{9}k = 1 \Rightarrow \frac{40}{9}k = 1 \Rightarrow k = \frac{9}{40}$

26)

$\frac{1}{125}k + \frac{1}{25}k + \frac{1}{5}k + k + k = 1 \Rightarrow \frac{281}{125}k = 1 \Rightarrow k = \frac{125}{281}$

$\mathrm{P}(X \leq 3) = \frac{1}{125} \times \frac{125}{281} + \frac{1}{25} \times \frac{125}{281} + \frac{1}{5} \times \frac{125}{281}$

$= \frac{1}{281} + \frac{5}{281} + \frac{25}{281} = \frac{31}{281}$

27)

$2k + 0.42 + 0.37 + k = 1 \Rightarrow 3k + 0.79 = 1$

$\Rightarrow k = 0.07$

$P(2 \times 3, 2 + 3) + P(3 \times 2, 3 + 2) + P(3 \times 3, 3 + 3)$

$= 0.37 \times 0.07 + 0.07 \times 0.37 + 0.07 \times 0.07 = 0.0567$

28)

$\mathrm{P}(1, 4) + \mathrm{P}(4, 1) + \mathrm{P}(2, 3) + \mathrm{P}(3, 2)$

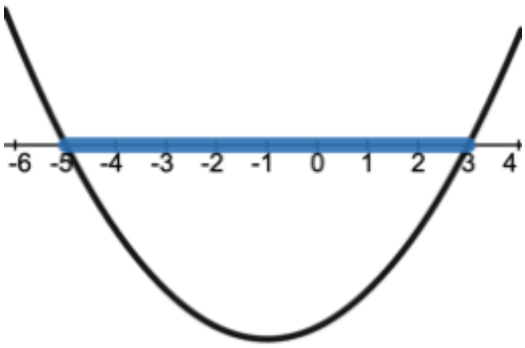
$= \frac{1}{6} \times \frac{5}{18} + \frac{5}{18} \times \frac{1}{6} + \frac{1}{3} \times \frac{1}{9} + \frac{1}{9} \times \frac{1}{3} = \frac{1}{6}$

29)

$\mathrm{P}(Y + 2X \leq 15)$ and $Y = X^2$

$\Rightarrow \mathrm{P}(X^2 + 2X \leq 15) = \mathrm{P}(X^2 + 2X - 15 \leq 0)$

$= \mathrm{P}((X + 5)(X - 3) \leq 0)$



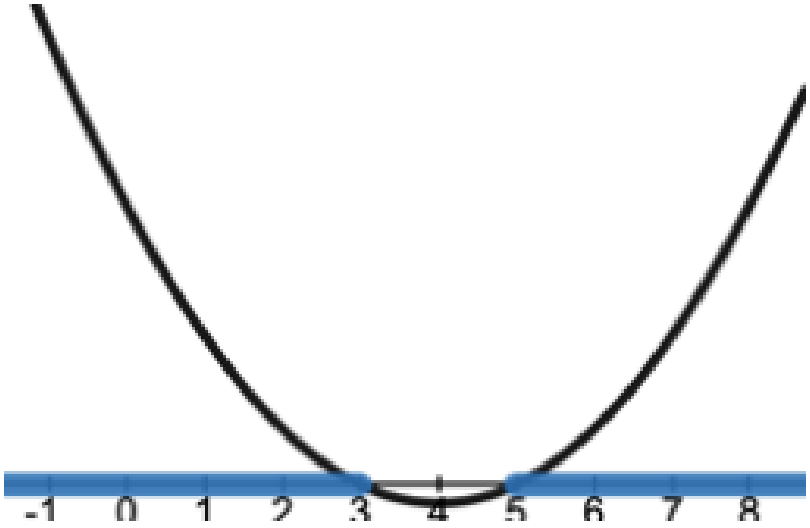
$= \mathrm{P}(2) + \mathrm{P}(3) = 0.89$

30)

$\mathrm{P}(X + Y \geq 8)$ and $Y = \frac{15}{X}$

$\Rightarrow \mathrm{P}\left(X + \frac{15}{X} \geq 8\right) = \mathrm{P}(X^2 + 15 \geq 8X)$

$= \mathrm{P}(X^2 - 8X + 15 \geq 0) = P((X - 3)(X - 5) \geq 0)$



$\mathrm{P}(5) + \mathrm{P}(9) + \mathrm{P}(13) = 0.2 + 0.05 + 0.05 = 0.3$

31 a)

$\mathrm{P}(0, 0, 0) + \mathrm{P}(1, 1, 1) + \mathrm{P}(2, 2, 2) + \mathrm{P}(3, 3, 3)$

$= \left(\frac{1}{15}\right)^3 + \left(\frac{1}{10}\right)^3 + \left(\frac{1}{3}\right)^3 + \left(\frac{1}{2}\right)^3 = \frac{49}{300}$

31 b)

$\mathrm{P}(0,0,2)+\mathrm{P}(0,2,0)+\mathrm{P}(2,0,0)+\mathrm{P}(1,1,0)+\mathrm{P}(1,0,1)+\mathrm{P}(0,1,1)$

$=3\times\left(\frac{1}{15}\right)^2\times\frac{1}{3}+3\times\left(\frac{1}{10}\right)^2\times\frac{1}{15}=\frac{29}{4500}$

31 c)

$\mathrm{P}(2,1,0)+\mathrm{P}(3,2,1)+\mathrm{P}(3,2,0)+\mathrm{P}(3,1,0)$

$=\frac{1}{3}\times\frac{1}{10}\times\frac{1}{15}+\frac{1}{2}\times\frac{1}{3}\times\frac{1}{10}+\frac{1}{2}\times\frac{1}{3}\times\frac{1}{15}+\frac{1}{2}\times\frac{1}{10}\times\frac{1}{15}$

$=\frac{1}{450}+\frac{1}{60}+\frac{1}{90}+\frac{1}{300}=\frac{1}{30}$

32)

$a+b+c+0.45=1\Rightarrow a+b+c=0.55\ (1)$

$\mathrm{P}(X\leq 2)=\frac{1}{3}\mathrm{P}(X\geq 3)$

$\Rightarrow a+b=\frac{1}{3}(c+0.45)\ (2)$

Substitute (2) into (1):

$\frac{1}{3}(c+0.45)+c=0.55\Rightarrow\frac{1}{3}c+0.15+c=0.55$

$\Rightarrow\frac{4}{3}c=0.4\Rightarrow c=0.3$

$\therefore\mathrm{P}(X\geq 3)=0.3+0.45=0.75$

33)

x	1	3	7
$\mathrm{P}(X=x)$	$\frac{k}{2}$	$\frac{k}{4}$	$\frac{k}{8}$

$\frac{k}{2}+\frac{k}{4}+\frac{k}{8}=1\Rightarrow\frac{7}{8}k=1\Rightarrow k=\frac{8}{7}$

x	1	3	7
$\mathrm{P}(X=x)$	$\frac{4}{7}$	$\frac{2}{7}$	$\frac{1}{7}$

$1-\left(\mathrm{P}(1+1\not>3)+\mathrm{P}(1+1\not>7)+\mathrm{P}(1+3\not>7)+\mathrm{P}(3+1\not>7)+P(3+3\not>7)\right)$

$=1-\left(\frac{4}{7}\times\frac{4}{7}\times\frac{2}{7}+\frac{4}{7}\times\frac{4}{7}\times\frac{1}{7}+\frac{4}{7}\times\frac{2}{7}\times\frac{1}{7}+\frac{2}{7}\times\frac{4}{7}\times\frac{1}{7}+\frac{2}{7}\times\frac{2}{7}\times\frac{1}{7}\right)$

$=1-\left(\frac{32}{343}+\frac{16}{343}+\frac{8}{343}+\frac{8}{343}+\frac{4}{343}\right)$

$=1-\frac{68}{343}=\frac{275}{343}$

34)

$3k+4k+5k+6k=1\Rightarrow 18k=1\Rightarrow k=\frac{1}{18}$

s	6	12	18	24
$\mathrm{P}(S=s)$	$\frac{1}{6}$	$\frac{2}{9}$	$\frac{5}{18}$	$\frac{1}{3}$

$\mathrm{P}(S_1+S_2=36)=\mathrm{P}(12+24=36)$

$+\mathrm{P}(24+12=36)+\mathrm{P}(18+18=36)$

$=\frac{2}{9}\times\frac{1}{3}+\frac{1}{3}\times\frac{2}{9}+\frac{5}{18}\times\frac{5}{18}=\frac{73}{324}$

35)

x	12	13
$\mathrm{P}(X=x)$	$\frac{12}{25}$	$\frac{13}{25}$

y	13	14
$\mathrm{P}(Y=y)$	$\frac{13}{27}$	$\frac{14}{27}$

$\mathrm{P}(X+Y=26)$
$= \mathrm{P}(12+14=26) + \mathrm{P}(13+13=26)$
$= \frac{12}{25} \times \frac{14}{27} + \frac{13}{25} \times \frac{13}{27} = \frac{337}{675}$

36)

x	1	2	3	4	5
$\mathrm{P}(X=x)$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$

$\mathrm{P}(X_1 = X_2) = \mathrm{P}(1=1) + \mathrm{P}(2=2) + \mathrm{P}(3=3)$
$+\mathrm{P}(4=4) + \mathrm{P}(5=5)$
$= \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2 = 5 \times \left(\frac{1}{5}\right)^2 = \frac{1}{5}$

37)

x	1	2	3	$\ldots$	n
$\mathrm{P}(X=x)$	$\frac{1}{n}$	$\frac{1}{n}$	$\frac{1}{n}$	$\ldots$	$\frac{1}{n}$

$\mathrm{P}(X_1 = X_2) = \mathrm{P}(1=1) + \mathrm{P}(2=2) + \mathrm{P}(3=3) + \cdots + \mathrm{P}(n=n)$
$= \left(\frac{1}{n}\right)^2 + \left(\frac{1}{n}\right)^2 + \left(\frac{1}{n}\right)^2 + \cdots + \left(\frac{1}{n}\right)^2$
$= n \times \left(\frac{1}{n}\right)^2 = \frac{1}{n}$

38)

As $\mathrm{P}(X_1 = X_2) = 0.1$, it must be that $\mathrm{P}(X_1 \neq X_2) = 0.9$. Half of these events will be $\mathrm{P}(X_1 < X_2)$ and the other half will be $\mathrm{P}(X_1 > X_2)$.

So $\mathrm{P}(X_1 < X_2) = \frac{0.9}{2} = 0.45$

39 a)

x	3	4	5	6
$\mathrm{P}(X=x)$	$6k$	$8k$	$10k$	$12k$

$6k + 8k + 10k + 12k = 1 \Rightarrow 36k = 1 \Rightarrow k = \frac{1}{36}$

x	3	4	5	6
$\mathrm{P}(X=x)$	$\frac{1}{6}$	$\frac{2}{9}$	$\frac{5}{18}$	$\frac{1}{3}$

$\mathrm{P}(3,6)+\mathrm{P}(6,3)+\mathrm{P}(4,5)+\mathrm{P}(5,4)$
$=\frac{1}{6}\times\frac{1}{3}+\frac{1}{3}\times\frac{1}{6}+\frac{2}{9}\times\frac{5}{18}+\frac{5}{18}\times\frac{2}{9}=\frac{19}{81}$

39 b)

$\mathrm{P}(X_1 \text{ or } X_2 \text{ is } 4 \mid X_1+X_2=9)=\frac{\mathrm{P}(X_1 \text{ or } X_2 \text{ is } 4\cap X_1+X_2=9)}{\mathrm{P}(X_1+X_2=9)}$
$=\frac{\mathrm{P}(4,5)+\mathrm{P}(5,4)}{\frac{19}{81}}=\frac{\frac{2}{9}\times\frac{5}{18}+\frac{5}{18}\times\frac{2}{9}}{\frac{19}{81}}=\frac{\frac{10}{81}}{\frac{19}{81}}=\frac{10}{19}$

40 a)

x	2	3	4	5
$\mathrm{P}(X=x)$	$\frac{4}{k}$	$\frac{8}{k}$	$\frac{16}{k}$	$\frac{32}{k}$

$\frac{4}{k}+\frac{8}{k}+\frac{16}{k}+\frac{32}{k}=1\Rightarrow\frac{60}{k}=1\Rightarrow k=60$

x	2	3	4	5
$\mathrm{P}(X=x)$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{4}{15}$	$\frac{8}{15}$

$\mathrm{P}(3+3+3=9)+\mathrm{P}(2+2+5=9)$
$+\mathrm{P}(2+5+2=9)+\mathrm{P}(5+2+2=9)$
$+\mathrm{P}(2+3+4=9)+\mathrm{P}(2+4+3=9)$
$+\mathrm{P}(3+2+4=9)+\mathrm{P}(3+4+2=9)$
$+\mathrm{P}(4+3+2=9)+\mathrm{P}(4+2+3=9)$
$=\left(\frac{2}{15}\right)^3+3\times\left(\frac{1}{15}\right)^2\times\frac{8}{15}+6\times\frac{1}{15}\times\frac{2}{15}\times\frac{4}{15}=\frac{16}{675}$

40 b)

$\mathrm{P}(\text{at least one } 4 \mid X_1+X_2+X_3=9)$
$=\frac{\mathrm{P}(\text{at least one } 4\cap X_1+X_2+X_3=9)}{\mathrm{P}(X_1+X_2+X_3=9)}=\frac{6\times\frac{1}{15}\times\frac{2}{15}\times\frac{4}{15}}{\frac{16}{675}}=\frac{\frac{16}{1125}}{\frac{16}{675}}=\frac{3}{5}$
This can also be spotted from there being 6 ways out of the 10 from above that 4 appeared, so $\frac{6}{10}=\frac{3}{5}$